

CURRICULUM VITAE



Miloš Vasić

AI Engineer · Software Engineer

LLM infrastructure, autonomous agents, and the governance that makes them trustworthy. Fleets over monoliths; anti-bluff, evidence-gated quality.

milos85vasic@gmail.com · milosvasic.ru · vasic.digital
GitHub: vasic-digital · HelixDevelopment · Server-Factory

2009

ENGINEERING SINCE

140+REPOSITORIES
GOVERNED**43**LLM PROVIDERS
UNIFIED**6+**

CORE LANGUAGES

AI Engineer — LLM infrastructure, autonomous agents, and the governance that makes them trustworthy.

- Email: milos85vasic@gmail.com
- Web: <https://milosvasic.ru> · <https://vasic.digital>
- GitHub: vasic-digital · HelixDevelopment · Server-Factory

Summary

AI/software engineer who builds AI-development systems end to end — from multi-provider LLM infrastructure and autonomous agents to the QA and governance layers that keep them honest. I architect fleets, not monoliths: large product applications on top of dozens of small, decoupled, independently-tested modules, all inheriting a shared engineering Constitution and verified by an anti-bluff, evidence-gated QA discipline. Primary language Go, with Kotlin/KMP, TypeScript/React, Python, Swift, and Shell. Guiding principle: a feature is done only when a real user can use it and there is captured evidence to prove it.

Core competencies

- **AI / LLM systems:** multi-provider LLM abstraction (40+ providers), MCP tool integration, RAG, vector databases & embeddings, agent orchestration (headless CLI agents, graph workflows, multi-round debate/consensus), planning (HiPlan/MCTS/Tree-of-Thoughts), LLMOps, benchmarking (SWE-bench/HumanEval/MMLU), LLM verification, defensive-LLM guardrails, computer-vision + LLM-vision.
- **Backend engineering:** Go (Gin), gRPC + Protobuf, HTTP/3 (QUIC), WebSockets, distributed systems (incl. TLA+ formal specs), high-throughput REST services and concurrent workers.
- **Data:** PostgreSQL, SQLite, SQLCipher (encryption at rest), Redis, Neo4j, ClickHouse, object storage (MinIO/S3/GCS/Azure).
- **Frontend / cross-platform:** TypeScript/React (Tailwind, Redux Toolkit, i18next), Angular, Electron, React Native, Kotlin Multiplatform, Android/Android TV (Kotlin), iOS (Swift), Tauri/Rust.
- **Infra / DevOps:** Docker & Compose, Kubernetes + Helm, Prometheus + Grafana, OpenTelemetry, QEMU/Libvirt/Parallels; CI/CD via GitHub Actions, Gradle, Make.
- **QA / quality engineering:** anti-bluff evidence-gated QA (HelixQA), Challenge harnesses with mutation gates, `go test -race`, visual-regression testing, ADB device testing, SonarQube, security scanning (semgrep/gosec/trivy/snyk/gitleaks/nancy).
- **Engineering governance:** Constitution-as-submodule; inheritance and propagation gates; documentation/coverage discipline across a 140+-repository fleet.

Selected projects

Governance & QA

- **HelixConstitution** — a universal engineering rulebook distributed as a Git submodule and inherited across 140+ repositories; mechanically-enforced anti-bluff law with propagation gates and mutation meta-tests.
- **HelixQA** — anti-bluff QA orchestration (Go): written test banks plus fully-autonomous LLM-and-vision QA sessions across Android/Android TV/Web/Desktop; no PASS without captured evidence.

AI development & LLM infrastructure

- **HelixAgent** — ensemble LLM service with multi-round model debate and verification-based provider selection.
- **HelixCode** — distributed AI development platform dividing work across SSH-managed workers with checkpoint/rollback.
- **HelixLLM** — single binary, six modes; OpenAI-/Anthropic-compatible inference over HTTP/3 with local llama.cpp and a scored fallback chain.
- **LLMProvider / LLMOrchestrator / LLMsVerifier** — one interface over 43 providers; a control plane for headless CLI agents; a verification source of truth with a mandatory model-comprehension gate.
- **HelixTrack** — a free-world JIRA alternative (flagship of the Helix-Track line).

Product-grade tools (vasic-digital utils)

- **Catalogizer** — multi-protocol, encrypted, self-hostable media collection management (Go/Gin + React), resilient to flaky network storage, built on 21 reusable submodules.
- **Courses-Creator** — markdown-to-video AI course pipeline with TTS and desktop/mobile/web players.
- **VisionEngine** — computer-vision + multi-provider LLM-vision UI perception with navigation graphs.
- **DocProcessor · Docs Chain · Herald · task_bridge · Vasic Digital Reusable Module Suite** — QA feature-mapping, content-hashed doc/DB sync, natural-language notifications, task/board sync, and the `digital.vasic.*` standard-library fleet.

Infrastructure automation (Server Factory)

- **Mail Server Factory** — declarative JSON → fully-provisioned, Dockerized mail servers across 12 connection types and 25 Linux distributions; reports 439 passing tests and a clean SonarQube gate.
- **Server Factory Core Framework, Qemu-Utills, Parallels-Utills** — the shared provisioning engine and VM-image tooling.

Languages & tools (quick list)

Go · Kotlin · Kotlin Multiplatform · TypeScript · JavaScript · Python · Swift · Java · Rust · Shell · PL/pgSQL · TLA+ · Gin · gRPC · HTTP/3 · React · Angular · Electron · React Native · PostgreSQL · SQLite · SQLCipher · Redis · Neo4j · ClickHouse · Docker · Kubernetes · Prometheus · Grafana · OpenTelemetry · QEMU · GitHub Actions · Gradle · Make

Experience

Software engineer since 2009, across the full development lifecycle — planning, development, team leading, and deployment. Full history below is sourced from the candidate's verified record (milosvasic.ru).

Full-time positions

- **SDK Developer — Harness** (harness.io), Belgrade, Serbia · 03/2020 – 12/2024. Lead developer on the family of SDKs for the company's Feature Flag division, focused on all major mobile platforms and beyond. Customers and partners included AWS, Google, and various banks. *Tech: Android, iOS, Flutter, React Native, TypeScript, JavaScript, Java, Kotlin, Swift, Go, Ruby.*
- **Software Engineer — Leica Geosystems** (leica-geosystems.com), Heerbrugg, Switzerland · 02/2016 – 02/2020. Primarily iOS and Android engineering for Leica Geosystems' cutting-edge 3D scanners — real-time communication with the hardware, data processing, and synchronization. Partner: Autodesk. *Tech: Android, iOS, Java, Kotlin, Swift, C++.*
- **SDK Developer — Bosch** (bosch.rs), Belgrade, Serbia · 01/2010 – 01/2016. Lead SDK developer for the Connected Vehicles SDK project — real-time Bluetooth communication with the OBD2 bus, high-performance data processing and persistence. *Tech: Android, Java, Kotlin.*

Other engagements

- **TN-TECH** (tn-tech.co.rs), Novi Sad, Serbia · part-time, from 03/2017. Work for Globex Data (Canada and Switzerland) — Sekur (SekurMessenger), SekurMail, SekurSuite — and the BusRide platform. *Tech: Android, Java, Kotlin, C++, Qt.*
- **Increment Loop** (incrementloop.com), Belgrade, Serbia · part-time, from 09/2023. The Yuno application. *Tech: Android, Kotlin.*
- **Open-source / own organizations** — HelixTrack, Server Factory (Mail Server Factory, Parallels-Utils, Qemu-Utils), and Vasic Digital (Android-Toolkit, Network-Binder), detailed under Selected projects above.

Publications

- **Fundamental Kotlin** — self-published author; latest revised edition September 2022 (Fundamental Kotlin, 3rd Edition). Also authored for Packt Publishing (UK).

Education

- **M.Sc, Contemporary Information Technologies** — University Singidunum, Belgrade, Serbia · 2014.
- **B.Sc, Informatics and Computing** — University Singidunum, Belgrade, Serbia · 2008.